গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
সিলেট-৩১১৪, বাংলাদেশ।
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Ref. no: CRTC/CEE/G20220981/S- 153

SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING
AT MAHMUDABAD R/A, KUMARGAON, AKHALIA, SYLHET.
DAG # SA-3260,3259, RS/BS-15923,15926, KHA # SA-67,639, RS/BS-15942,2096
J.L # SA-80, BS-70,64, MOUJA- KUMARGAON.

CLIENT

1. MOHAMMAD SHAIFUL ALAM AMIN
2. ANIMESH SARKAR
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5. MD. BAHUDDIN SIKDER
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CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTRD BY
M/S RELATION SOIL INVESTIGATION
Kumarpara, Sylhet.
Mob : 01711063018 & 01712186249

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-65 ft	13
02	Disturbed sample	02	0-70 ft	14
03	Disturbed sample	03	0-65 ft	13
04	Disturbed sample	04	0-65 ft	13
05	Disturbed sample	05	0-65 ft	13
06	Disturbed sample	06	0-65 ft	13
07	Disturbed sample	07	0-70 ft	14
08	Disturbed sample	08	0-65 ft	13
09	Disturbed sample	09	0-65 ft	13
10	Disturbed sample	10	0-65 ft	13
11	Disturbed sample	11	0-65 ft	13
12	Disturbed sample	12	0-65 ft	13

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		Tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	15	1.788	1.880	2.050	2.112	2.281	2.482	1.927
BH02	2.5-5.0	4	0.510	0.537	0.585	0.603	0.651	0.708	0.550
	5.0-10	14	1.668	1.754	1.912	1.970	2.128	2.315	1.797
BH03	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH04	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
BH05	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	12	1.392	1.464	1.596	1.644	1.776	1.932	1.500
BH06	2.5-5.0	4	0.418	0.439	0.479	0.493	0.533	0.580	0.450
	5.0-10	11	1.304	1.371	1.495	1.540	1.664	1.810	1.405
BH07	2.5-5.0	3	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
BH08	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH09	2.5-5.0	3	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	3	0.352	0.370	0.404	0.416	0.449	0.489	0.379
BH10	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH11	2.5-5.0	5	0.557	0.586	0.638	0.658	0.710	0.773	0.600
	5.0-10	15	1.708	1.796	1.958	2.017	2.179	2.371	1.841
BH12	2.5-5.0	3	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	11	1.312	1.380	1.504	1.550	1.674	1.821	1.414

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole											
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10	BH11	BH12
5	0.049	0.049	0.049	0.049	0.049	0.049	0.049	0.049	0.049	0.049	0.049	0.049
10	0.074	0.074	0.049	0.049	0.049	0.096	0.074	0.049	0.049	0.049	0.074	0.049
15	0.174	0.156	0.049	0.049	0.049	0.096	0.096	0.074	0.118	0.074	0.174	0.118
20	0.272	0.256	0.074	0.049	0.248	0.200	0.049	0.049	0.049	0.049	0.256	0.208
25	0.208	0.136	0.074	0.261	0.200	0.248	0.096	0.118	0.074	0.138	0.120	0.174
30	0.074	0.156	0.096	0.096	0.096	0.138	0.118	0.156	0.049	0.074	0.096	0.118
35	0.049	0.118	0.118	0.138	0.096	0.096	0.049	0.174	0.096	0.074	0.049	0.074
40	0.049	0.138	0.096	0.074	0.074	0.156	0.049	0.096	0.096	0.096	0.074	0.074
45	0.074	0.118	0.118	0.118	0.096	0.138	0.096	0.118	0.118	0.118	0.074	0.096
50	0.200	0.156	0.200	0.200	0.200	0.200	0.160	0.200	0.200	0.200	0.200	0.762
55	0.208	0.296	0.749	0.208	0.200	0.208	0.192	0.208	0.200	0.216	0.208	0.669
60	0.224	0.288	1.297	0.184	0.762	0.216	0.736	0.869	0.200	1.231	0.184	0.669
65	0.264	0.184	0.192	0.192	0.896	0.280	0.762	0.669	0.224	1.271	0.184	0.669
70	-	0.176	-	-	-	-	0.208	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole											
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10	BH11	BH12
5	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450
10	0.675	0.675	0.450	0.450	0.450	0.900	0.675	0.450	0.450	0.450	0.675	0.450
15	1.800	1.575	0.450	0.450	0.450	0.900	0.900	0.675	1.125	0.675	1.800	1.125
20	54.400	51.200	0.675	0.450	49.600	40.000	0.450	0.450	0.450	0.450	51.200	41.600
25	41.600	27.200	0.675	2.700	40.000	49.600	0.900	1.125	0.675	1.350	24.000	1.800
30	0.675	1.575	0.900	0.900	0.900	1.350	1.125	1.575	0.450	0.675	0.900	1.125
35	0.450	1.125	1.125	1.350	0.900	0.900	0.450	1.800	0.900	0.675	0.450	0.675
40	0.450	1.350	0.900	0.675	0.675	1.575	0.450	0.900	0.900	0.900	0.675	0.675
45	0.675	1.125	1.125	1.125	0.900	1.350	0.900	1.125	1.125	1.125	0.675	0.900
50	40.000	1.575	40.000	40.000	40.000	40.000	32.000	40.000	40.000	40.000	40.000	12.825
55	41.600	59.200	12.600	41.600	40.000	41.600	38.400	41.600	40.000	43.200	41.600	11.250
60	44.800	57.600	21.825	36.800	12.825	43.200	12.375	14.625	40.000	20.700	36.800	11.250
65	52.800	36.800	38.400	38.400	15.075	56.000	12.825	11.250	44.800	21.375	36.800	11.250
70	-	35.200	-	-	-	-	41.600	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole											
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10	BH11	BH12
5	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032
10	0.048	0.048	0.032	0.032	0.032	0.062	0.048	0.032	0.032	0.032	0.048	0.032
15	0.113	0.101	0.032	0.032	0.032	0.062	0.062	0.048	0.077	0.048	0.113	0.077
20	0.177	0.166	0.048	0.032	0.161	0.130	0.032	0.032	0.032	0.032	0.166	0.135
25	0.135	0.088	0.048	0.170	0.130	0.161	0.062	0.077	0.048	0.090	0.078	0.113
30	0.048	0.101	0.062	0.062	0.062	0.090	0.077	0.101	0.032	0.048	0.062	0.077
35	0.032	0.077	0.077	0.090	0.062	0.062	0.032	0.113	0.062	0.048	0.032	0.048
40	0.032	0.090	0.062	0.048	0.048	0.101	0.032	0.062	0.062	0.062	0.048	0.048
45	0.048	0.077	0.077	0.077	0.062	0.090	0.062	0.077	0.077	0.077	0.048	0.062
50	0.130	0.101	0.130	0.130	0.130	0.130	0.104	0.130	0.130	0.130	0.130	0.496
55	0.135	0.192	0.487	0.135	0.130	0.135	0.125	0.135	0.130	0.140	0.135	0.435
60	0.146	0.187	0.843	0.120	0.496	0.140	0.478	0.565	0.130	0.800	0.120	0.435
65	0.172	0.120	0.125	0.125	0.582	0.182	0.496	0.435	0.146	0.826	0.120	0.435
70	-	0.114	-	-	-	-	0.135	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole											
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10	BH11	BH12
5	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.203	0.203
10	0.304	0.304	0.203	0.203	0.203	0.405	0.304	0.203	0.203	0.203	0.304	0.203
15	0.810	0.709	0.203	0.203	0.203	0.405	0.405	0.304	0.506	0.304	0.810	0.506
20	24.480	23.040	0.304	0.203	22.320	18.000	0.203	0.203	0.203	0.203	23.040	18.720
25	18.720	12.240	0.304	1.215	18.000	22.320	0.405	0.506	0.304	0.608	10.800	0.810
30	0.304	0.709	0.405	0.405	0.405	0.608	0.506	0.709	0.203	0.304	0.405	0.506
35	0.203	0.506	0.506	0.608	0.405	0.405	0.203	0.810	0.405	0.304	0.203	0.304
40	0.203	0.608	0.405	0.304	0.304	0.709	0.203	0.405	0.405	0.405	0.304	0.304
45	0.304	0.506	0.506	0.506	0.405	0.608	0.405	0.506	0.506	0.506	0.304	0.405
50	18.000	0.709	18.000	18.000	18.000	18.000	14.400	18.000	18.000	18.000	18.000	5.771
55	18.720	26.640	5.670	18.720	18.000	18.720	17.280	18.720	18.000	19.440	18.720	5.063
60	20.160	25.920	9.821	16.560	5.771	19.440	5.569	6.581	18.000	9.315	16.560	5.063
65	23.760	16.560	17.280	17.280	6.784	25.200	5.771	5.063	20.160	9.619	16.560	5.063
70	-	15.840	-	-	-	-	18.720	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed six-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is irregular in between the boreholes and vertical inconsistency exists in the soil formation.

In the case of boreholes BH 01, BH 02, BH 05, BH 06, BH11 and BH12, soft clay soil with low bearing capacity is observed at the top 15 ft of the EGL followed by a dense sandy layer in the subsequent 5 ft to 10 ft. In the case of other boreholes (BH 03, BH 04, BH 07, BH 08, BH 09 and BH 10), soft clay soil with low bearing capacity is seen up to a depth of 20 to 25 ft from the EGL. An organic layer (15 to 20 ft thick) exists between 30 ft to 45 ft of EGL in the case of all boreholes (BH 01 to BH 12). Dense sand or hard clay soil is observed from 50ft of the EGL in all boreholes except BH 02. In BH 02, dense sand with good bearing capacity is begun from 55 ft of the EGL.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since soft clay soil with low bearing capacity is found in the top 15 ft layer and a thick layer of organic soil exists underneath, shallow foundation is unlikely to be a safe option in the existing condition. However, the bearing capacity for the shallow foundation in the existing soil condition is given in **Table 7**.

Considering the site soil condition, deep foundation can be preferentially recommended to ensure the stability of the structure. The size and depth of the pile should be selected in the light of bearing capacity given in the report (Table 8-11).

Safety provisions

Since the vertical soil profile of the proposed site confirmed the existence of organic and cohesive soil, it is recommended to check the **settlement** of the site considering the building load. For the pile foundation, P**ile load test** covering the entire project area should be carried out for confirming the design parameters of the service pile. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundations.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. The client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **65-70** ft as client provided.

Report Prepared By	**Test Assisted By**	**Counter Signed By**
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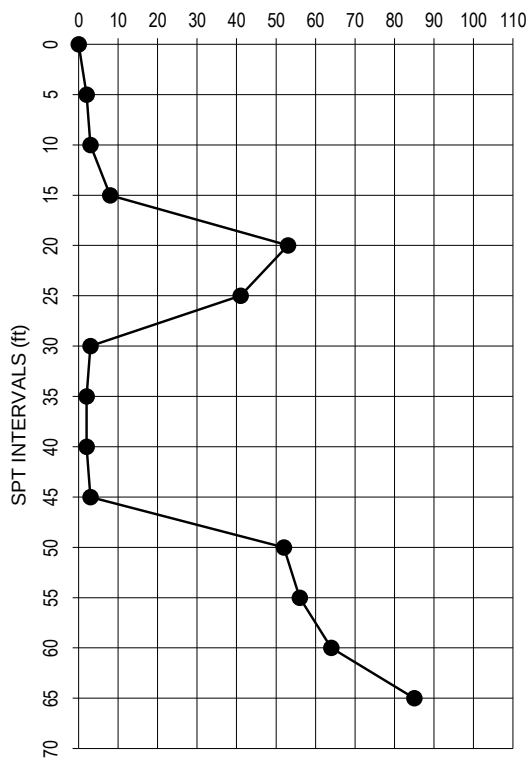
RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **01**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 02/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 02/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	10.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	2	3	
D3		15.0	5.0	Grey sandy clay		15.0	3	4	4	8	
D4		20.0	10.0	Grey fine sand		20.0	12	26	27	53	
D5		25.0				25.0	14	20	21	41	
D6		30.0	15.0	Black Organic		30.0	1	1	2	3	L.C-26'
D7		35.0				35.0	1	1	1	2	
D8		40.0				40.0	1	1	1	2	
D9		45.0	5.0	Grey soft clay		45.0	1	1	2	3	
D10		50.0	20.0	Grey fine sand		50.0	20	25	27	52	L.C-47'
D11		55.0				55.0	22	26	30	56	
D12		60.0				60.0	27	30	34	64	
D13		65.0				65.0	30	35	50	85	

GRAPHICAL REPRESANTATION OF N- VALUES

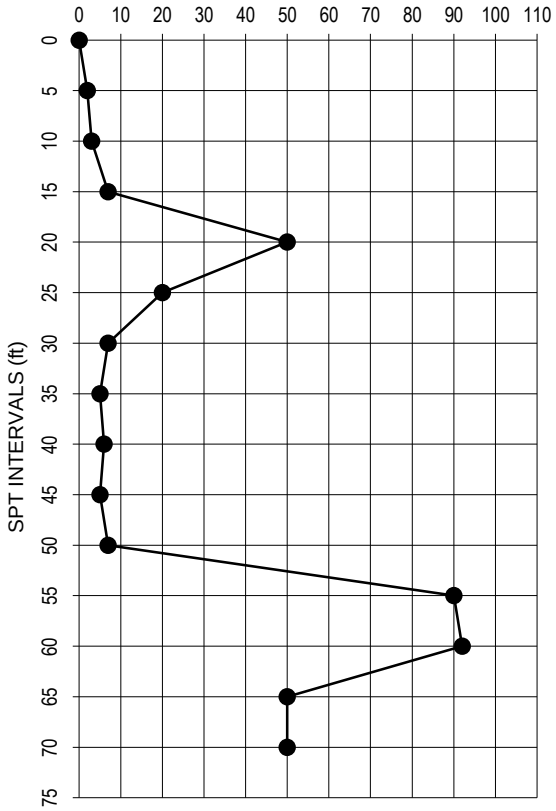


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.		
BORE HOLE NO. **02**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 02/08/2022		
BORING DIA : **0.33 ft**			Sand with silt & clay.
DEPTH OF BORING : 70ft	COMPLETION DATE : 02/08/2022		
			Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	10.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	2	3	
D3		15.0	5.0	Grey sandy clay		15.0	2	3	4	7	
D4		20.0	10.0	Grey fine sand		20.0	17	23	27	50	
D5		25.0				25.0	10	13	7	20	
D6		30.0	20.0	Black Organic		30.0	2	3	4	7	L.C-23'
D7		35.0				35.0	2	2	3	5	
D8		40.0				40.0	2	3	3	6	
D9		45.0				45.0	2	2	3	5	
D10		50.0	5.0	Grey sandy clay		50.0	3	3	4	7	
D11		55.0	20.0	Grey fine sand		55.0	30	40	50	90	L.C-52'
D12		60.0				60.0	32	42	50	92	
D13		65.0				65.0	41	50	over	50	
D14		70.0				70.0	45	50	over	50	

GRAPHICAL REPRESANTATION OF N- VALUES

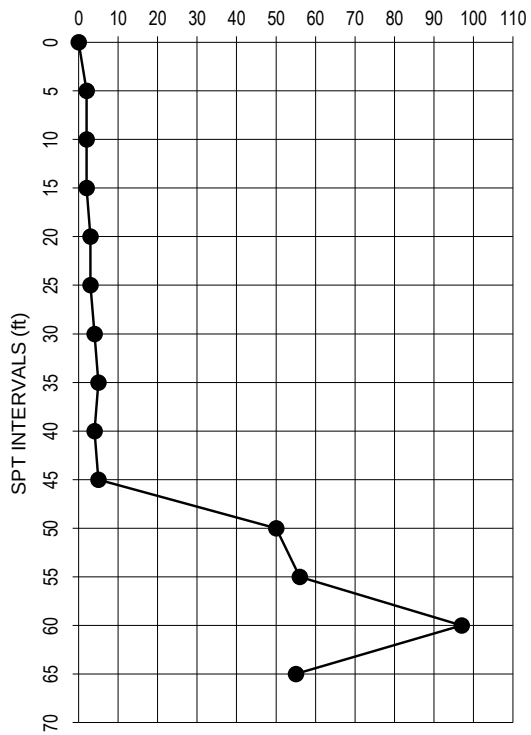


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **03**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 03/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 03/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	25.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	1	1	2	
D4		20.0				20.0	1	1	2	3	
D5		25.0				25.0	1	1	2	3	
D6		30.0	15.0	Black Organic		30.0	1	2	2	4	
D7		35.0				35.0	2	2	3	5	
D8		40.0				40.0	2	2	2	4	
D9		45.0	5.0	Grey soft clay		45.0	2	2	3	5	
D10		50.0	5.0	Grey fine sand		50.0	20	24	26	50	L.C-49'
D11		55.0	10.0	Grey hard clay		55.0	22	26	30	56	
D12		60.0				60.0	29	47	50	97	
D13		65.0	5.0	Grey fine sand		65.0	50	55	over	55	

GRAPHICAL REPRESANTATION OF N- VALUES

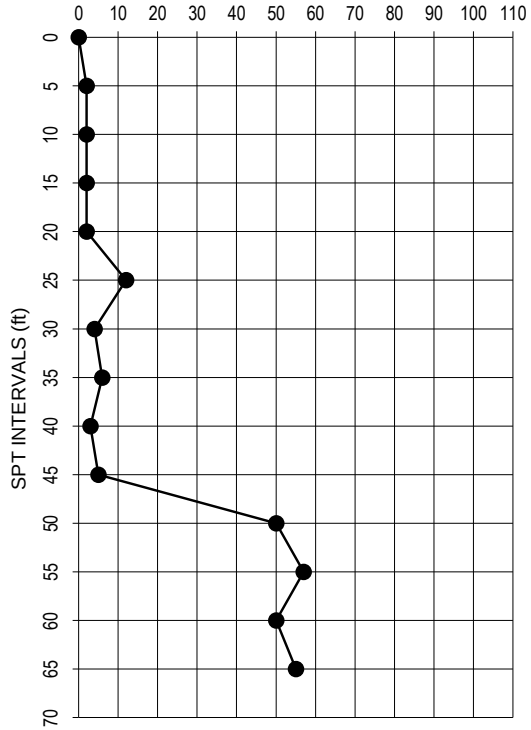


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **04**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 03/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 03/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	20.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	1	1	2	
D4		20.0				20.0	1	1	1	2	
D5		25.0	5.0	Grey stiff clay		25.0	2	10	2	12	
D6		30.0	15.0	Black Organic		30.0	1	2	2	4	
D7		35.0				35.0	2	3	3	6	
D8		40.0				40.0	2	2	1	3	
D9		45.0	5.0	Grey sandy clay		45.0	2	3	2	5	
D10		50.0	20.0	Grey fine sand		50.0	14	23	27	50	L.C-48'
D11		55.0				55.0	22	27	30	57	
D12		60.0				60.0	44	50	over	50	
D13		65.0				65.0	47	55	over	55	

GRAPHICAL REPRESANTATION OF N- VALUES

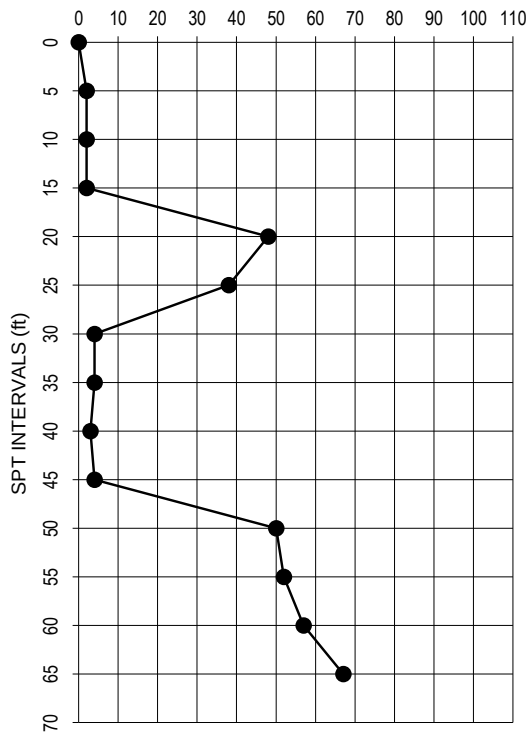


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **05**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 04/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 04/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	15.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	1	1	2	
D4		20.0	10.0	Grey fine sand		20.0	12	22	26	48	
D5		25.0				25.0	10	18	20	38	
D6		30.0	15.0	Black Organic		30.0	2	2	2	4	
D7		35.0				35.0	2	2	2	4	
D8		40.0				40.0	2	1	2	3	
D9		45.0	5.0	Grey sandy clay		45.0	2	2	2	4	
D10		50.0	10.0	Grey fine sand		50.0	16	23	27	50	L.C-48'
D11		55.0				55.0	21	25	27	52	
D12		60.0	10.0	Grey hard clay		60.0	24	27	30	57	
D13		65.0				65.0	27	32	35	67	

GRAPHICAL REPRESANTATION OF N- VALUES

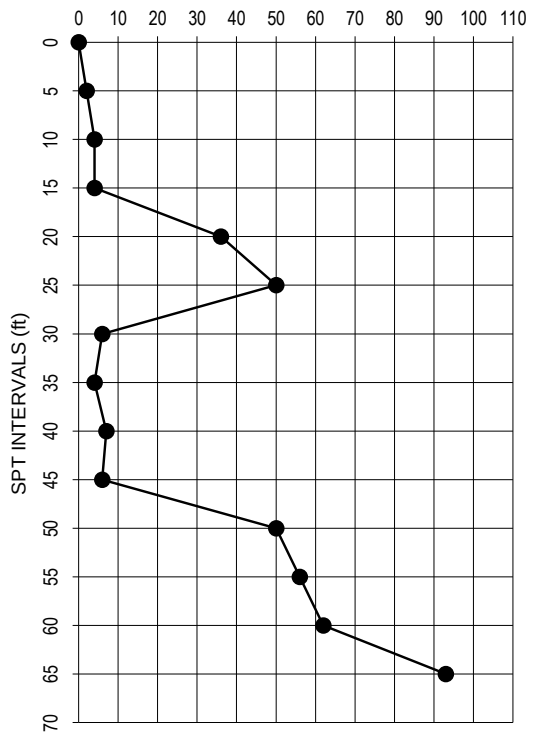


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **06**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 04/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 04/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	15.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	2	2	2	4	
D3		15.0				15.0	2	2	2	4	
D4		20.0	10.0	Grey fine sand		20.0	7	15	21	36	
D5		25.0				25.0	18	24	26	50	
D6		30.0	15.0	Black Organic		30.0	2	2	4	6	L.C-26'
D7		35.0				35.0	2	2	2	4	
D8		40.0				40.0	3	3	4	7	
D9		45.0	5.0	Grey sandy clay		45.0	2	3	3	6	
D10		50.0	20.0	Grey fine sand		50.0	17	23	27	50	L.C-48'
D11		55.0				55.0	22	26	30	56	
D12		60.0				60.0	28	30	32	62	
D13		65.0				65.0	32	43	50	93	

GRAPHICAL REPRESANTATION OF N- VALUES

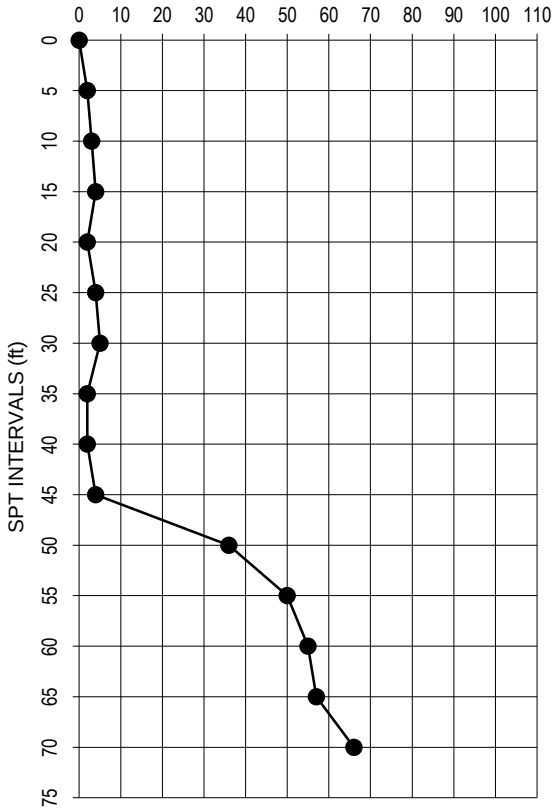


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.		
BORE HOLE NO. **07**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 04/08/2022		
BORING DIA : **0.33 ft**			Sand with silt & clay.
DEPTH OF BORING : 70ft	COMPLETION DATE : 04/08/2022		
			Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	20.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	2	3	
D3		15.0				15.0	2	2	2	4	
D4		20.0				20.0	1	1	1	2	
D5		25.0	15.0	Black Organic		25.0	1	2	2	4	
D6		30.0				30.0	2	2	3	5	
D7		35.0				35.0	1	1	1	2	
D8		40.0	10.0	Grey soft clay		40.0	2	1	1	2	
D9		45.0				45.0	2	2	2	4	
D10		50.0	10.0	Grey fine sand		50.0	10	15	21	36	L.C-48'
D11		55.0				55.0	14	23	27	50	
D12		60.0	10.0	Grey hard clay		60.0	20	25	30	55	
D13		65.0				65.0	24	27	30	57	
D14		70.0	5.0	Grey fine sand		70.0	27	32	34	66	

GRAPHICAL REPRESANTATION OF N- VALUES

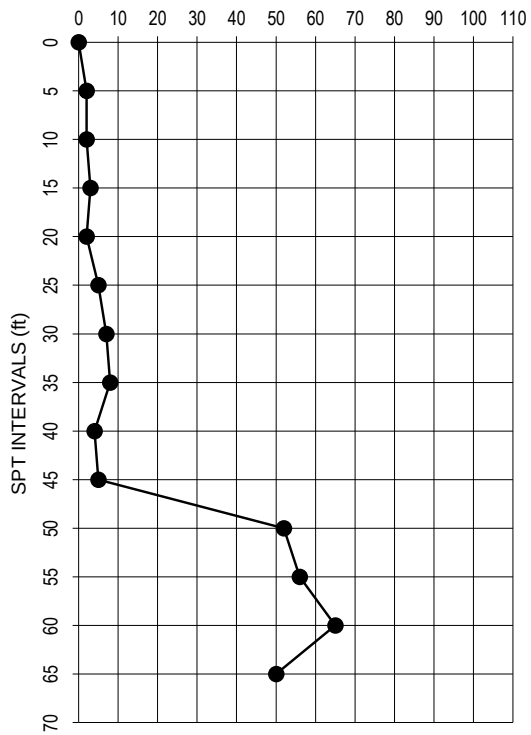


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **08**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 05/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 05/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	25.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	1	2	3	
D4		20.0				20.0	1	1	1	2	
D5		25.0				25.0	2	2	3	5	
D6		30.0	15.0	Black Organic		30.0	2	3	4	7	
D7		35.0				35.0	3	3	5	8	
D8		40.0				40.0	3	2	2	4	
D9		45.0	5.0	Grey soft clay		45.0	2	2	3	5	
D10		50.0	10.0	Grey fine sand		50.0	22	25	27	52	L.C-47'
D11		55.0				55.0	25	26	30	56	
D12		60.0	10.0	Grey hard clay		60.0	27	30	35	65	
D13		65.0				65.0	42	50	over	50	

GRAPHICAL REPRESANTATION OF N- VALUES

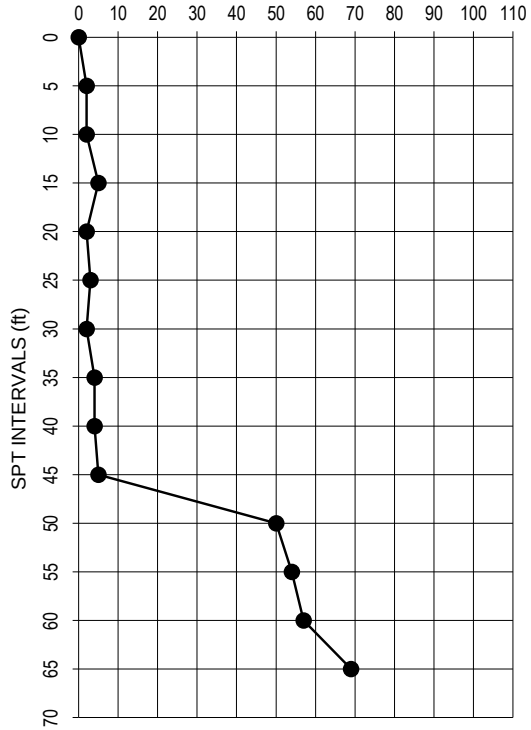


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **09**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 05/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 05/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	25.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	2	2	3	5	
D4		20.0				20.0	1	1	1	2	
D5		25.0				25.0	1	2	1	3	
D6		30.0	15.0	Black Organic		30.0	1	1	1	2	
D7		35.0				35.0	1	2	2	4	
D8		40.0				40.0	1	2	2	4	
D9		45.0	5.0	Grey sandy clay		45.0	2	2	3	5	
D10		50.0	20.0	Grey fine sand		50.0	17	24	26	50	L.C-47'
D11		55.0				55.0	19	26	28	54	
D12		60.0				60.0	22	27	30	57	
D13		65.0				65.0	27	32	37	69	

GRAPHICAL REPRESANTATION OF N- VALUES

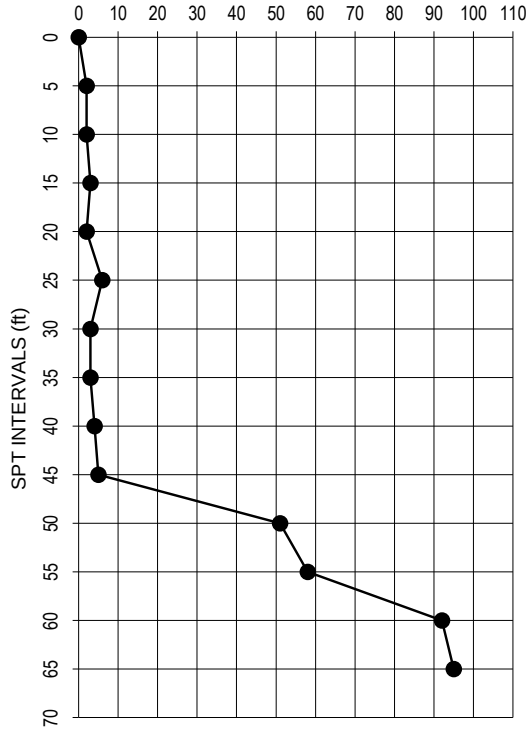


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **10**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 05/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 05/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	25.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	1	2	3	
D4		20.0				20.0	1	1	1	2	
D5		25.0				25.0	2	4	2	6	
D6		30.0	15.0	Black Organic		30.0	1	1	2	3	
D7		35.0				35.0	1	1	2	3	
D8		40.0				40.0	1	2	2	4	
D9		45.0	5.0	Grey soft clay		45.0	2	2	3	5	
D10		50.0	10.0	Grey fine sand		50.0	21	24	27	51	L.C-47'
D11		55.0				55.0	22	27	31	58	
D12		60.0	10.0	Grey hard clay		60.0	27	42	50	92	
D13		65.0				65.0	35	45	50	95	

GRAPHICAL REPRESANTATION OF N- VALUES

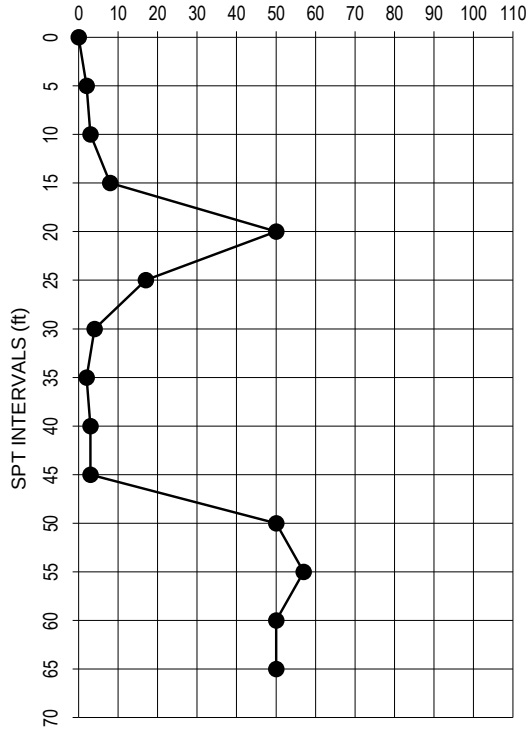


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **11**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 06/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 06/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	10.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	2	3	
D3		15.0	5.0	Grey sandy clay		15.0	2	3	5	8	
D4		20.0	10.0	Grey fine sand		20.0	15	23	27	50	L.C-17'
D5		25.0				25.0	13	10	7	17	L.C-23'
D6		30.0	20.0	Black Organic		30.0	1	2	2	4	
D7		35.0				35.0	1	1	1	2	
D8		40.0				40.0	1	1	2	3	
D9		45.0				45.0	1	1	2	3	
D10		50.0	20.0	Grey fine sand		50.0	13	24	26	50	L.C-48'
D11		55.0				55.0	18	25	32	57	
D12		60.0				60.0	35	50	over	50	
D13		65.0				65.0	35	50	over	50	

GRAPHICAL REPRESANTATION OF N- VALUES

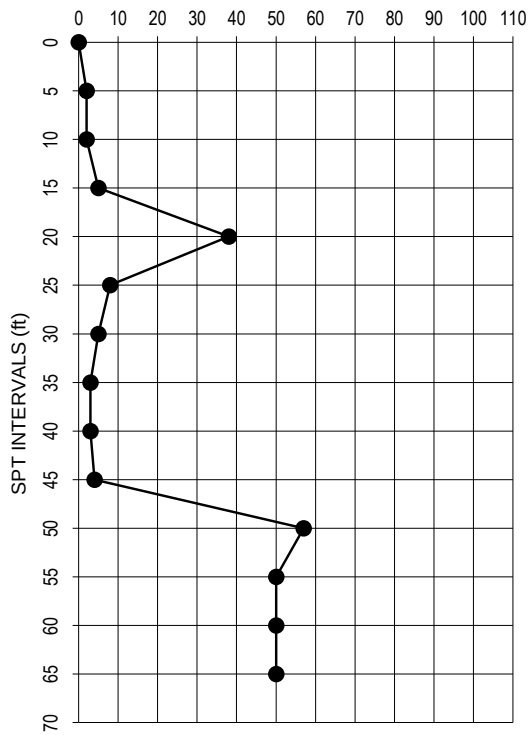
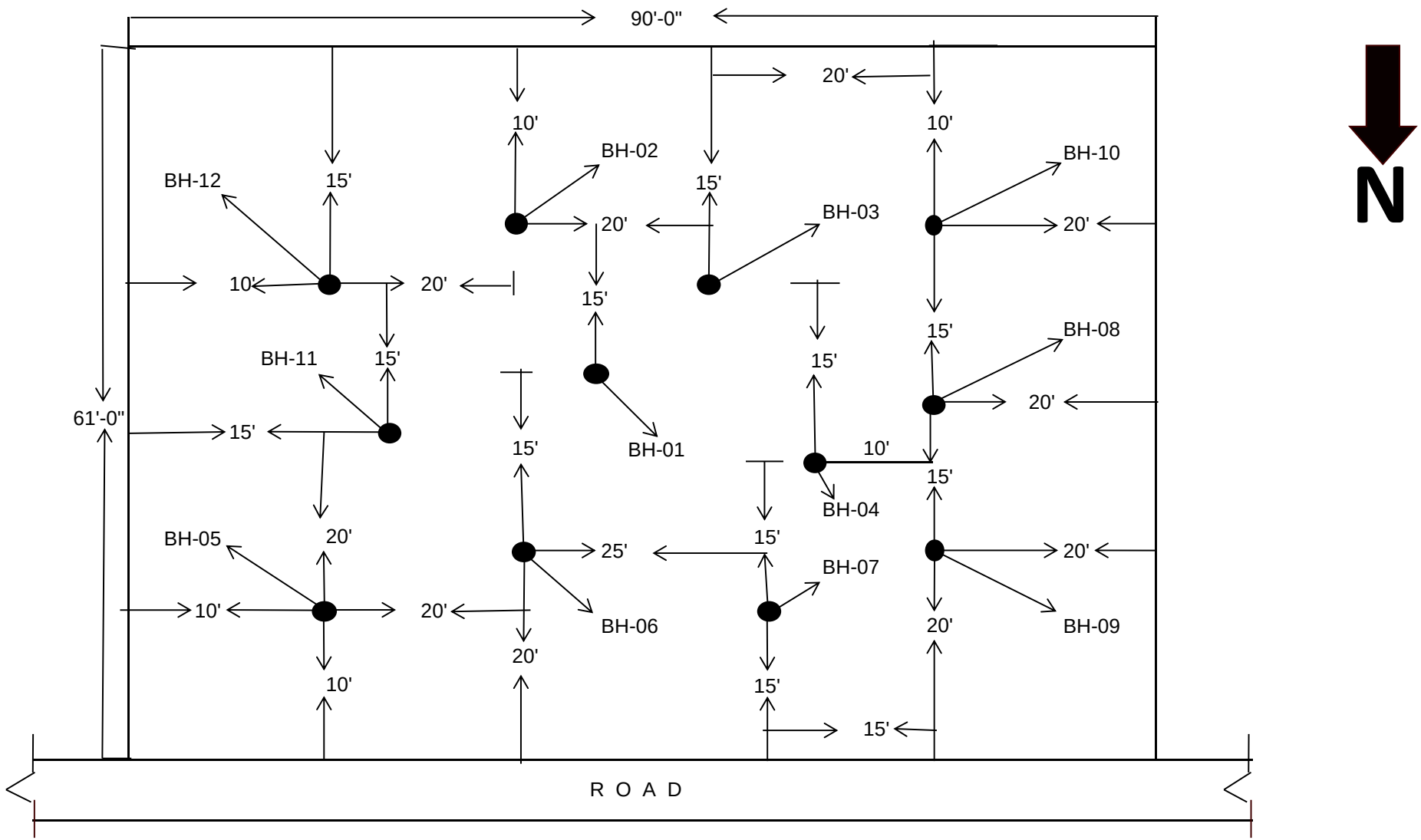


RECORD OF BORING AND TESTING

CLIENT : MOHAMMAD SHAIFUL ALAM AMIN & GONG	LOCATION: Mahmudabad R/A, Akhalia, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 06 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 0'-0" BELOW EGL.	
BORE HOLE NO. **12**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 06/08/2022	
BORING DIA : **0.33 ft**		Sand with silt & clay.
DEPTH OF BORING : 65ft	COMPLETION DATE : 06/08/2022	
		Organic/coarse sand

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value	Layer Change
						0	0	0	0	0	
D1		5.0	15.0	Grey soft clay		5.0	1	1	1	2	
D2		10.0				10.0	1	1	1	2	
D3		15.0				15.0	1	2	3	5	
D4		20.0	5.0	Grey fine sand		20.0	12	18	20	38	L.C-18'
D5		25.0	20.0	Black Organic		25.0	2	3	5	8	
D6		30.0				30.0	2	2	3	5	
D7		35.0				35.0	1	1	2	3	
D8		40.0				40.0	1	1	2	3	
D9		45.0	5.0	Grey soft clay		45.0	1	2	2	4	
D10		50.0	20.0	Grey hard clay		50.0	25	27	30	57	L.C-47'
D11		55.0				55.0	30	50	over	50	
D12		60.0				60.0	50	50	over	50	
D13		65.0				65.0	50	50	over	50	

GRAPHICAL REPRESANTATION OF N- VALUES



CRTC, CEE DEPARTMENT, SUST SYLHET	SITE PLAN(NOT IN SCALE)	**PROJECT NAME :** Construction of 06 Storied Residential building. **LOCATION :** Mahmudabad R/A, Akhalia, Sylhet.